

● HARD

● ALGEBRA

● ANSWER KEY

Linear functions

02

10 answers · Rationale-rich · Teach from doc

Q1**B****B. $(-4,0)$**

Choice B is correct. The equation of a linear function can be written in the form $y = mx + b$, where $y = f(x)$, m is the slope of the graph of $y = f(x)$, and b is the y-coordinate of the y-intercept of the graph. The value of m can be found using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. According to the table, the points $(-11, 21)$ and $(-10, 18)$ lie on the graph of $y = f(x)$. Using these two points in the slope formula yields $m = \frac{18 - 21}{-10 - (-11)}$, or -3 . Substituting -3 for m in the slope-intercept form of the equation yields $y = -3x + b$. The value of b can be found by substituting values from the table and solving; for example, substituting the coordinates of the point $(-11, 21)$ into the equation $y = -3x + b$ gives $21 = -3(-11) + b$, which yields $b = -12$. This means the function given by the table can be represented by the equation $y = -3x - 12$. The value of the x-intercept of the graph of $y = f(x)$ can be determined by finding the value of x when $y = 0$. Substituting $y = 0$ into $y = -3x - 12$ yields $0 = -3x - 12$, or $x = -4$. This corresponds to the point $(-4, 0)$.

Choice A is incorrect and may result from substituting the value of the slope for the x-coordinate of the x-intercept. Choice C is incorrect and may result from a calculation error. Choice D is incorrect and may result from substituting the y-coordinate of the y-intercept for the x-coordinate of the x-intercept.

Q2

A

A. $f(x) = 60x + 100$

Choice A is correct. It's given that the window repair specialist charges \$ 220 for the first two hours of repair plus an hourly fee for each additional hour. Let n represent the hourly fee for each additional hour after the first two hours. Since it's given that x is the number of hours of repair, it follows that the charge generated by the hourly fee after the first two hours can be represented by the expression $nx - 2$. Therefore, the total cost, in dollars, for x hours of repair is

$fx = 220 + nx - 2$. It's given that the total cost for 5 hours of repair is \$ 400. Substituting 5 for x and 400 for fx into the equation $fx = 220 + nx - 2$ yields $400 = 220 + n5 - 2$, or $400 = 220 + 3n$. Subtracting 220 from both sides of this equation yields $180 = 3n$. Dividing both sides of this equation by 3 yields $n = 60$. Substituting 60 for n in the equation

$fx = 220 + nx - 2$ yields $fx = 220 + 60x - 2$, which is equivalent to $fx = 220 + 60x - 120$, or $fx = 60x + 100$. Therefore, the total cost, in dollars, for x hours of repair is $fx = 60x + 100$.

Choice B is incorrect. This function represents the total cost, in dollars, for x hours of repair where the specialist charges \$ 340, rather than \$ 220, for the first two hours of repair.

Choice C is incorrect. This function represents the total cost, in dollars, for x hours of repair where the specialist charges \$ 160, rather than \$ 220, for the first two hours of repair, and an hourly fee of \$ 80, rather than \$ 60, after the first two hours.

Choice D is incorrect. This function represents the total cost, in dollars, for x hours of repair where the specialist charges \$ 380, rather than \$ 220, for the first two hours of repair, and an hourly fee of \$ 80, rather than \$ 60, after the first two hours.

Q3**A**

A. $P = \frac{w}{110}$

Choice A is correct. It's given that w represents the total wall area, in square feet. Since the walls of the room will be painted twice, the amount of paint, in gallons, needs to cover $2w$ square feet. It's also given that one gallon of paint will cover 220 square feet. Dividing the total area, in square feet, of the surface to be painted by the number of square feet covered by one gallon of paint gives the number of gallons of paint that will be needed. Dividing $2w$ by 220 yields $\frac{2w}{220}$, or $\frac{w}{110}$. Therefore, the equation that represents the total amount of paint P , in gallons, needed to paint the walls of the room twice is $P = \frac{w}{110}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from finding the amount of paint needed to paint the walls once rather than twice.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**C**

C. $\frac{61c}{4} + 105$

Choice C is correct. It's given that $gc + 7 = \frac{c}{4}$. Therefore, for the given linear function g , when $x = c + 7$, $gx = \frac{c}{4}$. Substituting $c + 7$ for x and $\frac{c}{4}$ for gx in $gx = b - 15x$ yields $\frac{c}{4} = b - 15c + 7$. Applying the distributive property to the right-hand side of this equation yields $\frac{c}{4} = b - 15c - 105$. Adding $15c$ to both sides of this equation yields $\frac{c}{4} + 15c = b - 105$. Adding 105 to both sides of this equation yields $\frac{c}{4} + 15c + 105 = b$, or $\frac{61c}{4} + 105 = b$. Therefore, the expression that represents the value of b is $\frac{61c}{4} + 105$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5**A**

A. $y = 215x + 215$

Choice A is correct. It's given that the cost of renting a large canopy tent is \$ 430 for the first day and \$ 215 for each additional day for up to 10 days. For x days of renting the tent, the cost includes \$ 430 for the first day and \$ 215 for each of the $(x - 1)$ additional days. It follows that the cost y , in dollars, of renting the tent can be expressed as $y = 430 + 215(x - 1)$, which is equivalent to $y = 430 + 215x - 215$, or $y = 215x + 215$. Therefore, the equation $y = 215x + 215$ gives the cost of renting the tent for x days, where x is a positive integer and $x \leq 10$.

Choice B is incorrect. This equation represents a situation where the cost of renting the tent for the first day is \$ 215, not \$ 430, and the cost for each additional day is \$ 430, not \$ 215.

Choice C is incorrect. This equation represents a situation where the cost of renting the tent for the first day is \$ 645, not \$ 430, and the cost for each additional day is \$ 430, not \$ 215.

Choice D is incorrect. This equation represents a situation where the cost of renting the tent for the first day is \$ 645, not \$ 430.

Q6**D**

D. The average price per gallon on December 1, 2014

Choice D is correct. Since 2.74 is a constant term, it represents an actual price of gas rather than a measure of change in gas price. To determine what gas price it represents, find x such that $F(x) = 2.74$, or $2.74 = 2.74 - 0.19(x - 3)$. Subtracting 2.74 from both sides gives $0 = -0.19(x - 3)$. Dividing both sides by -0.19 results in $0 = x - 3$, or $x = 3$. Therefore, the average price of gas is \$2.74 per gallon 3 months after September 1, 2014, which is December 1, 2014.

Choice A is incorrect. Since 2.74 is a constant, not a multiple of x , it cannot represent a rate of change in price. Choice B is incorrect. The difference in the average price from September 1, 2014, to December 1, 2014, is $F(3) - F(0) = 2.74 - 0.19(3 - 3) - (2.74 - 0.19(0 - 3)) = 2.74 - (2.74 + 0.57) = -0.57$, which is not 2.74. Choice C is incorrect. The average price per gallon on September 1, 2014, is $F(0) = 2.74 - 0.19(0 - 3) = 2.74 + 0.57 = 3.31$, which is not 2.74.

Q7**A****A.** 16.38

Choice A is correct. It's given that the function $Fx = \frac{9}{5}x - 273.15 + 32$ gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. A temperature that increased by 9.10 kelvins means that the value of x increased by 9.10 kelvins. It follows that an increase in x by 9.10 increases $F(x)$ by $\frac{9}{5}9.10$, or 16.38. Therefore, if a temperature increased by 9.10 kelvins, the temperature increased by 16.38 degrees Fahrenheit.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**-12**

The correct answer is -12. It's given that the functions f and g are defined as $fx = \frac{1}{4}x - 9$ and $gx = \frac{3}{4}x + 21$. If the function h is defined as $hx = fx + gx$, then substituting $\frac{1}{4}x - 9$ for fx and $\frac{3}{4}x + 21$ for gx in this function yields $hx = \frac{1}{4}x - 9 + \frac{3}{4}x + 21$. This can be rewritten as $hx = \frac{4}{4}x + 12$, or $hx = x + 12$. The x -intercept of a graph in the xy -plane is the point on the graph where $y = 0$. The equation representing the graph of $y = hx$ is $y = x + 12$. Substituting 0 for y in this equation yields $0 = x + 12$. Subtracting 12 from both sides of this equation yields $-12 = x$, or $x = -12$. Therefore, the x -coordinate of the x -intercept of the graph of $y = hx$ in the xy -plane is -12.

Q9**A**

A. $f(n) = 14n + 175$

Choice A is correct. A tour group with n people, where $n \geq 25$, can be split into two subgroups: the first 25 people and the additional $n - 25$ people. Since the museum charges \$ 21 per person for the first 25 people and \$ 14 for each additional person, the charge for the first 25 people is \$ 21(25) and the charge for the additional $n - 25$ people is \$ 14($n - 25$). Therefore, the total charge, in dollars, is given by the function $f(n) = 21(25) + 14(n - 25)$, or $f(n) = 14n + 175$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10**D**

D. $y = 135x + 135$

Choice D is correct. It's given that the cost of renting a backhoe for up to 10 days is \$ 270 for the first day and \$ 135 for each additional day. Therefore, the cost y , in dollars, for x days, where $x \leq 10$, is the sum of the cost for the first day, \$ 270, and the cost for the additional $x - 1$ days, \$ 135 $x - 1$. It follows that $y = 270 + 135x - 1$, which is equivalent to $y = 270 + 135x - 135$, or $y = 135x + 135$.

Choice A is incorrect. This equation represents a situation where the cost of renting a backhoe is \$ 135 for the first day and \$ 270 for each additional day.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank